

How I learned to tolerate the telescope

Remote observations of ion tails of comets

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Comets in the solar wind

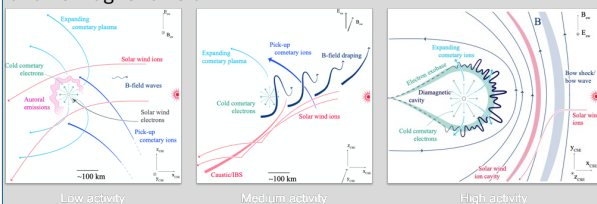
Comets are small icy rocks that originate from the edges of our solar system. They are leftovers of solar system formation and therefore of interest to the scientific community.

Comets are distinguished from asteroids by their outgassing behaviour. As the comet moves into the inner solar system its surface is heated by the Sun and the ices sublimate.

This forms a neutral coma which is then ionized.

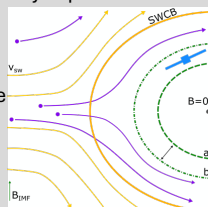
The main ions in comets are H_2O^+ , H_3O^+ , CO^+ , and CO_2^+ .

The ion cloud around a comet is an obstacle to the solar wind and its magnetic field.



The environment is interesting due to its variety of processes:

ion-neutral interactions, ion gyroradius effects, chemical reactions, and interaction with dust. All of these processes change as the comet becomes more active and different regions form depending on gas production rate. A field free region can form around the nucleus as the solar wind and its magnetic field are expelled.



Rosetta

The Rosetta mission explored comet 67P/Churyumov-Gerasimenko for over two years and accompanied the comet through an entire perihelion passage. It made it possible to observe the development of regions and interaction surfaces in the environment.

When interpreting the data from Rosetta, one of the main constraints is the lack of reliable solar wind measurements at the comet.

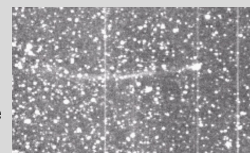
Comet Interceptor

Comet Interceptor is the next mission to a comet. Like Giotto in 1986, it will perform a single fly-by of an as yet unknown target. CI will for the first time observe the magnetic field from three individual spacecraft. However, only some solar wind context will be provided during the flyby.

Tail dynamics from optical telescopes

Telescopes have observed comets forever. Many findings are incidental, for example the observations by the STEREO spacecraft have been invaluable to characterize the tails of near-Sun comets. The Ulysses spacecraft also crossed a cometary tail, giving invaluable measurements of ions of cometary origin at 1AU.

Particularly striking events are so called disconnection events, where the tail seems to completely separate from the nucleus. These events are well observed but poorly understood.



Conclusions

Comets are a unique plasma laboratory that can teach us about complex, multi-ion plasmas in the solar wind. The combination of simulations, in-situ measurements, and optical observations is a significant step forward in understanding this environment.

Simulations

Much was learned from Rosetta, but without the context from the solar wind and multi-point measurements, further interpretations remain elusive.

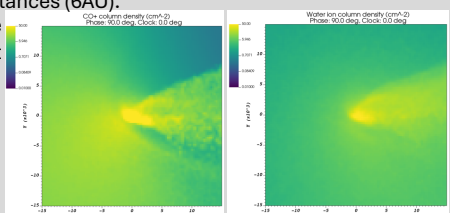
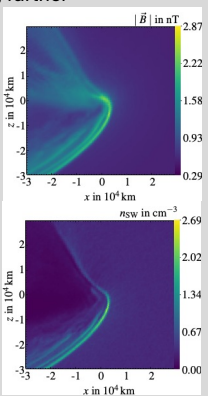
Simulations can provide context for the measurements and in the future link in-situ measurements to remote observations.

The heavy ions as comets have very large gyroradii in the solar wind, which makes ion-kinetic Hybrid simulations the tool of choice for this environment.

The AIKEF code has been used successfully in the past to understand the cometary plasma environment.

As an example, 29P/Schwassmann-Wachmann is a well observed but puzzling comet that outgasses at unusually high heliocentric distances (6AU).

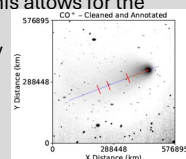
Simulations are set up to reflect the measured properties of the neutral gas coma, and ions are produced according to photoionization rates and charge exchange. The integrated line of sight density can then be directly compared to observations.



Narrowband imaging

Narrowband filters can be used to isolate the emissions from the ions. To assess the images quantitatively, another image of the dust continuum is necessary to isolate the ion emissions.

Narrowband imaging can be done quite fast at bright comets, with exposure times of less than 1 minute. This allows for the investigation of tail dynamics at short time-scales, but at the cost of quantitative density measurements as a dust continuum image is required for those.



VLT/MUSE

MUSE is an integral field unit which measures all wavelengths within its capability at the same time. This has the advantage of giving a precise spectrum of the light emitted or reflected by the comet and makes it possible to isolate the dust continuum at the same time as providing quantitative estimates of the ion emissions. However, long exposure times (10s of minutes) are needed to achieve good signal to noise. This means cometary tails features are smeared out and only long-term variations can be investigated. Nevertheless this is a useful tool that has recently given observations of the ion tail of the interstellar object 3I/Atlas.

